

Q. Represent (1259.125) in IEEE double precision format.

$$1259.125$$

$$100111010111.001$$

$$1.00111010111001 \times 2^{10}$$

$$e - 1023 = 10$$

$$e = 1033 \Rightarrow 10000001001$$

$$[0 | 10000001001 | 0011101011100100]$$

Unit-2

Boolean algebra :-

It is nothing but the mathematics of digital system.

LANS -

(1) Associate law :-

- (a) $A + (B + C) = (A + B) + C$
- (b) $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

(2) Commutative law :-

$$A + B = B + A$$

$$A \cdot B = B \cdot A$$

(3) Distributive law :-

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

$$A \cdot (B + C) = (A \cdot B) + (A \cdot C)$$

Figure 3.1 Data science process model.

Theorem 1

i) Theorem 1

$$A + 0 = A$$

i/p	o/p
A	$A + 0 = Y$
1	$1 + 0 = 1$
0	$0 + 0 = 0$

Theorem 2

$$A + A = A$$

A	$A + A = Y = A$
0	$0 + 0 = 0$
1	$1 + 1 = 1$

Theorem 3

$$A + 1 = 1$$

A	$A + 1 = Y$
0	$0 + 1 = 1$
1	$1 + 1 = 1$

Theorem 4

$$A \cdot A = A$$

A	$A \cdot A = Y = A$
0	$0 \cdot 0 = 0 = 0$
1	$1 \cdot 1 = 1$

Theorem 5

$$A \cdot 0 = 0$$

A	$A \cdot 0 = Y$
0	$0 \cdot 0 = 0$
1	$1 \cdot 0 = 0$

Theorem 6

$$A + \bar{A} = 1$$

A	$\bar{A}$	$A + \bar{A} = Y$
0	1	$0 + 1 = 1$
1	0	$1 + 0 = 1$

Theorem 7

$$A \cdot 1 = A$$

A	$A \cdot 1 = A$
0	$0 \cdot 1 = 0$
1	$1 \cdot 1 = 1$

Theorem 8

$$A \cdot \bar{A} = 0$$

A	$\bar{A}$	$A \cdot \bar{A}$
0	1	$0 \cdot 1 = 0$
1	0	$1 \cdot 0 = 0$

Theorem 9

$$\bar{\bar{A}} = A$$

Spiral

In order to work with massive data sets, orgs, should have the capability to process large qty

calc, data processing & reasoning

A	$\bar{A}$	$\bar{\bar{A}} = \bar{A}$
0	1	0
1	0	1

Theorem 10

$$A + AB = A$$

Proof $A + AB = A(1+B)$
 $= A \cdot 1 (1+B=1)$
 $= A$

Theorem 11

$$A + \bar{A}B = A + B$$

$$\begin{aligned} A + \bar{A}B &= (A + AB) + \bar{A}B \quad (A + AB = A) \\ &= A \cdot A + AB + \bar{A}B \quad (A \cdot A = A) \\ &= A \cdot A + AB + A \cdot \bar{A} + \bar{A}B \quad (A \cdot \bar{A} = 0) \\ &= A(A + B) + \bar{A}(A + B) \\ &= (A + B)(A + \bar{A}) \\ &= (A + B) \cdot 1 \\ &= A + B \end{aligned}$$

Theorem 12

$$A + BC = (A + B)(A + C)$$

$$\begin{aligned} \text{L.H.S.} &= A + BC = A \cdot 1 + BC \quad (A \cdot 1 = A) \\ &= A(1+B) + BC \\ &= A \cdot 1 + A \cdot B + BC \\ &= A(1+C) + AB + BC \\ &= A + AC + AB + BC \end{aligned}$$

$$\begin{aligned}
 &= A \cdot A + AC + AB + BC \\
 &= A(A+C) + B(A+C) \\
 &= (A+B)(A+C) \\
 &\quad \text{etc.}
 \end{aligned}$$

De - Morgan's theorem:

- Theorem-1: $\overline{A+B} = \overline{A} \cdot \overline{B}$

- Theorem-2: $\overline{A \cdot B} = \overline{A} + \overline{B}$

Theorem-1:

It states that the comp. of sum of 2 or more variables is equivalent to the product of comp. of each variables.

Theorem-2:

It states that the comp. of product of 2 or more variables is equi. to the sum of the comp. of each variables.

Proof:

for theorem 1:

$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

from boolean expression

$$A + \overline{A} = 1$$

$$A \cdot \overline{A} = 0$$

So we can prove

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$$\begin{aligned}
 \text{L.H.S.} &= (A+B) + (\overline{A+B}) \\
 &= (A+B) + \overline{A \cdot B} \\
 &= (A+B + \overline{A})(A+B + \overline{B}) \\
 &= (\overline{A+A} + B)(\overline{A+B} + \overline{B}) \\
 &= (\overline{1+B})(\overline{A+1}) \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 &A+B \cdot C \\
 &= (A+B)(A+C)
 \end{aligned}$$

Similarly:

$$\begin{aligned}
 \text{L.H.S.} &= (A+B) \cdot (\overline{A+B}) \\
 &= (A+B) \cdot \overline{A \cdot B} \\
 &= (\overline{A \cdot A} \cdot \overline{B}) + \overline{B \cdot A \cdot B} \\
 &= 0
 \end{aligned}$$

2. Proof of Demorgan's 2nd theorem:

$$\overline{A \cdot B} = \overline{A} \cdot \overline{B}$$

By Boolean exp-

$$A + \overline{A} = 1$$

$$A \cdot \overline{A} = 0$$

So we can prove:

$$(a). (A \cdot B) + (\overline{A \cdot B}) = 1$$

$$(b). (A \cdot B) \cdot (\overline{A \cdot B}) = 0$$

$$(a) (A \cdot B) + (\overline{A \cdot B}) = 1$$

L.H.S.

$$(A \cdot B) + (\overline{A \cdot B})$$

$$= (A \cdot B) + (\overline{A+B})$$

$$= (\overline{A+B} + A) \cdot (\overline{A+B} + B)$$

$$A+B \cdot C = (A+B)(A+C)$$

$$= (A + A + B)(\bar{A} + \bar{B} + B)$$

$$= (1 + B)(\bar{A} + 1)$$

$$= 1 \quad \text{R.H.S.}$$

(b) $(A \cdot B) \cdot (A \cdot \bar{B}) = 0$

L.H.S. $= (A \cdot B)(A \cdot \bar{B})$

$$= (A \cdot B)(\bar{A} + \bar{B})$$

$$= A \cdot B \cdot \bar{A} + A \cdot B \cdot \bar{B}$$

$$= 0 \quad \text{R.H.S.}$$

Q. Solve $\overline{A + BC}$ using De Morgan's theorem and also implement using basic gates.

$$\overline{A + BC}$$

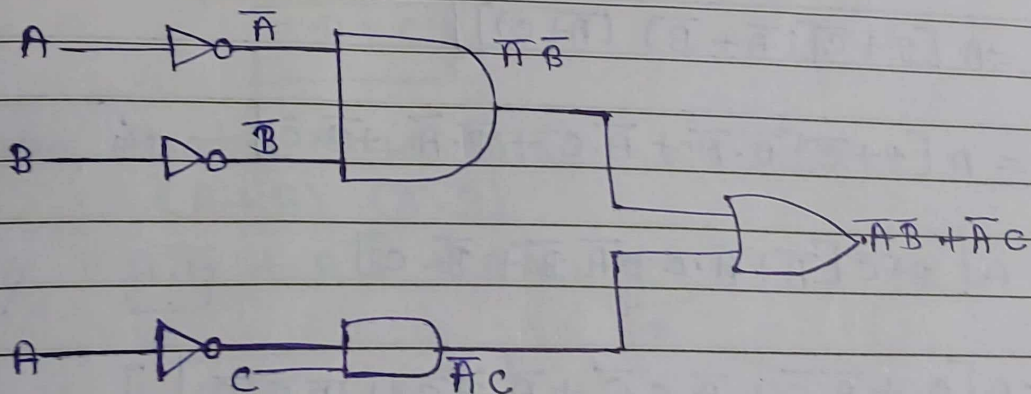
$$= \bar{A}(\bar{B} + \bar{C})$$

$$= \bar{A}[\bar{B} + \bar{C}]$$

$$= \bar{A} \cdot \bar{B} + \bar{A} \cdot \bar{C}$$

$$(\overline{A + B} = \bar{A} \cdot \bar{B})$$

$$(\overline{A \cdot B} = \bar{A} + \bar{B})$$



Q. Reduce the given boolean function.

$$(\bar{A} + A + B)(\bar{B} + B + C)$$

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In order to work with massive data sets, org., should have the capability to store & process large qty of data.

Algo. can perform cal., data processing & reasoning tasks.

...line or algo. the

$$= (\overline{A + A + B}) + (\overline{B + B + C})$$

$$= (\overline{A + A \cdot B}) + (\overline{B + B \cdot C})$$

$$= \overline{A} \cdot (1 + \overline{B}) + \overline{B} \cdot (1 + \overline{C})$$

$$= \overline{A} + \overline{B}$$

$$\boxed{= A + B}$$

Q. $A [B + \overline{C} [AB + A\overline{C}]]$

$$= A [B + \overline{C} [AB + A\overline{C}]]$$

$$= A [B + \overline{C} [\overline{AB} \cdot \overline{A\overline{C}}]]$$

$$= A [B + \overline{C} [(\overline{A + B})(\overline{A + \overline{C}})]]$$

$$= A [B + \overline{C} [(\overline{A + B})(\overline{A + C})]]$$

$$= A [B + \overline{C} [\overline{A} \cdot \overline{A} + \overline{A} \cdot \overline{C} + \overline{B} \cdot \overline{A} + \overline{B} \cdot \overline{C}]]$$

$$= A [B + \overline{C} [\overline{A} + \overline{A} \cdot \overline{C} + \overline{A} \overline{B} + \overline{B} \cdot \overline{C}]]$$

$$= A [B + \overline{A} \overline{C} + \overline{A} \cdot \overline{C} \cdot \overline{C} + \overline{A} \overline{B} \overline{C} + \overline{B} \overline{C} \overline{C}]$$

$$= A [B + \overline{A} \overline{C} + \overline{A} \overline{B} \overline{C}]$$

$$= A \cdot B + \underbrace{A \overline{A} \overline{C}}_0 + \underbrace{A \overline{A} \overline{B} \overline{C}}_0$$

Spiral

$$[= AB]$$

Q. Reduce the given boolean expression.

$$A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C$$

$$\begin{aligned} & A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C \\ &= A\bar{B}C + B + B\bar{D}(1+A) + \bar{A}C \\ &= A\bar{B}C + B + B\bar{D} + \bar{A}C \\ &= C(A\bar{B} + \bar{A}) + B(1+\bar{D}) \\ &= C[\bar{A} + A(\bar{A}+B)] + B \\ &= C[\bar{A} + B] + B \\ &= \bar{A}C + \bar{B}C + B \\ &= \bar{A}C + B + \bar{B}C \\ &= \bar{A}C + [(B+\bar{B})(B+C)] \\ &= \bar{A}C + [B+C] \\ &= \bar{A}C + B + C \\ &= C(1+\bar{A}) + B \\ &[= B + C] \end{aligned}$$

$$\begin{aligned} & A + BC \\ &= (A+B)(A+C) \\ &= (\bar{A}+A)(\bar{A}+B) \\ &= C(\bar{A}+B) + B \end{aligned}$$

Q. Prove that $(A+B)(\bar{A}\cdot\bar{B}) = A\cdot\bar{B} + B\cdot\bar{A}$

$$(A+B)(\bar{A}\cdot\bar{B})$$

$$\underbrace{A\cdot\bar{A}}_0 + A\cdot\bar{B} + B\cdot\bar{A} + \underbrace{B\cdot\bar{B}}_0$$

$$[= A\cdot\bar{B} + B\cdot\bar{A}]$$

Q. Prove that $\overline{AB+AC} = \bar{A}\cdot\bar{B}\cdot\bar{C} = \bar{A} + \bar{B}\cdot\bar{C}$

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Algo. can perform cal., data processing & reasoning task

$$\begin{aligned}
&= A\bar{B} \cdot \bar{A}C = \bar{A} \cdot B \cdot \bar{C} \\
&= (\bar{A} + B)(\bar{A} + \bar{C}) + \bar{A}B\bar{C} \\
&= \bar{A} \cdot \bar{A} + \bar{A}\bar{C} + \bar{A}B + B\bar{C} + \bar{A}B\bar{C} \\
&= \bar{A} + \bar{A}\bar{C} + \bar{A}B(1 + \bar{C}) + B\bar{C} \\
&= \bar{A} + \bar{A}\bar{C} + \bar{A}B + B\bar{C} \\
&= \bar{A}(1 + \bar{C}) + B(\bar{A} + \bar{C}) \\
&= \bar{A} + \bar{A}\bar{B} + B\bar{C} \\
&= \bar{A}(1 + \bar{B}) + B\bar{C} \\
&= \boxed{\bar{A} + B\bar{C}}
\end{aligned}$$

Q. $\overline{A\bar{B} + ABC} + A(B + A\bar{B})$

$$\begin{aligned}
&= \overline{A\bar{B} + ABC} \cdot A(B + A\bar{B}) \\
&= (\overline{A\bar{B} + ABC}) [\bar{A} + (\bar{B} + A\bar{B})] \\
&= (\overline{A\bar{B} + ABC}) [\bar{A} + \bar{B} \cdot \bar{A}\bar{B}] \\
&= (\overline{A\bar{B} + ABC}) [\bar{A} + \bar{B}(\bar{A} \cdot \bar{B})] \\
&= (\overline{A\bar{B} + ABC}) (\bar{A} + \bar{B}[\bar{A} \cdot \bar{B}]) \\
&= (\overline{A\bar{B} + ABC}) (\bar{A} + \bar{B} \cdot \bar{A} + \bar{B} \cdot \bar{B}) \\
&= 0
\end{aligned}$$

Q. $[A\bar{B}(C + BD) + \bar{A} \cdot \bar{B}]c = \bar{B}c$

L.H.S.

$$\begin{aligned}
&= [A\bar{B}(C + BD) + \bar{A} \cdot \bar{B}]c \\
&= A\bar{B}C \cdot c + 0 + \bar{A} \cdot C \cdot \bar{B}c \\
&= A\bar{B}c + \bar{A}C \cdot \bar{B}c \\
&= c[A\bar{B} + \bar{A} \cdot \bar{B}] \\
&= c\bar{B}[A + \bar{A}] \\
&= \bar{B}c
\end{aligned}$$

Q. Draw the logical diagram for the given boolean exp.

$$Y = AB + AC + A \cdot B \cdot C$$

L.H.S.

$$= AB + AC + A \cdot B \cdot C$$

$$= (AB + (A)(B+C)) + A \cdot B \cdot C$$

$$= (A+B)(A+C) + A \cdot B \cdot C$$

$$= A + B + C + A \cdot B \cdot C$$

$$A(1+B \cdot C) + B + C = A + B + C$$

$$= AB + A + C + A \cdot B \cdot C$$

$$= B(A + A \cdot C) + A + C$$

$$= B(A \cdot A)(A + C) + A + C$$

$$= B(A + C) + A + C$$

$$= AB + B \cdot C + A + C$$

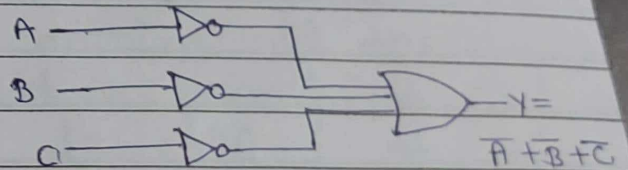
$$= AB + C(1+B) + A$$

$$= AB + C + A$$

$$= A + AB + C$$

$$= (A+A)(A+B) + C$$

$$= A + B + C$$



Q. Implement the following funcⁿ using NAND gates only.

$$Y = (A + BC)(B + CA)$$

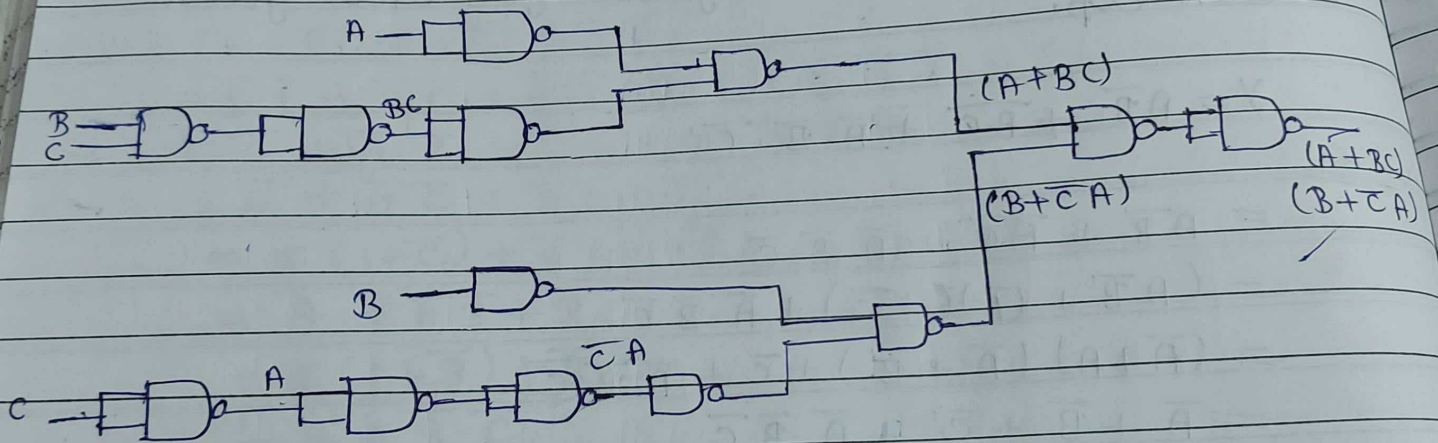
$$= AB + A \cdot AC + B \cdot BC + ABC \cdot C$$

$$= AB + AC + BC + 0$$

Spiral

In order to
work with massive data sets,
stability

Algo. can perform
cal., data processing
& reasoning tasks



Sum of product (SOP):

more than a product sum of 2 or sum of product (SOP) sum terms, is called weeds, we can say that it is OR'ing of AND operated variables.

Ex: $\overline{A}B + ABC + \overline{A}\overline{B}\overline{C}$
 $= \overline{A}B\overline{C} + \overline{A}BC + \overline{A}\overline{B}\overline{C}$

- SOP form is ~~retain~~ ^{written} where func. is 1.

Minterm:

$0 \rightarrow \overline{A}$

$1 \rightarrow A$

- If a boolean func. contains 'n' variables, then a product term which contains all the 'n' variables once in either complemented or non-complemented form is called the min-term.

- For 'n' variable

No. of minterm = 2^n (For n variables)

For A, B, C (3 variables)

No. of minterm = $2^3 = 8$

Spiral

In order to work with massive data sets, org., should have the capability to store & process large qty

Algo. can perform cal., data processing & reasoning task

Decimal	A	B	C	Minterm	Minterm
0	0	0	0	m_0	$\overline{A} \overline{B} \overline{C}$
1	0	0	1	m_1	$\overline{A} \overline{B} C$
2	0	1	0	m_2	$\overline{A} B \overline{C}$
3	0	1	1	m_3	$\overline{A} B C$
4	1	0	0	m_4	$A \overline{B} \overline{C}$
5	1	0	1	m_5	$A \overline{B} C$
6	1	1	0	m_6	$A B \overline{C}$
7	1	1	1	m_7	$A B C$

#	A	B	C	O/P
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

$$Y = \overline{A} \overline{B} C + \overline{A} B \overline{C} + A \overline{B} \overline{C} + A \overline{B} C + A B C$$

↓
std. canonical form

$$Y(A, B, C) = m_1 + m_2 + m_4 + m_5 + m_7$$

$$Y(A, B, C) = \sum_m (1, 2, 4, 5, 7)$$

Q. Convert the following boolean exp. into canonical, SOP form.

$$F(A, B, C) = (A + BC)(B + CA)$$

$$F(A, B, C) = (A + BC)(B + CA)$$

$$= AB + AAC + B \cdot B \cdot C + BC \cdot CA$$

$$= AB + AC + BC + \cancel{ABC} + \cancel{ABC}$$

$$= AB(C + \bar{C}) + AC(B + \bar{B}) + BC(A + \bar{A})$$

$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}B\bar{C}$$

$$= ABC + AB\bar{C} + A\bar{B}C + \bar{A}BC + \bar{A}B\bar{C} + \bar{A}\bar{B}C$$

$$Y(A, B, C) = \sum m(3, 4, 6, 7)$$

Q. $F(A, B, C, D) = ABCD + \bar{A}BC + \bar{B}\bar{C}$

$$= ABCD + \bar{A}BC(D + \bar{D}) + \bar{B}\bar{C}(A + \bar{A})(D + \bar{D})$$

$$= ABCD + \bar{A}BCD + \bar{A}BC\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$= ABCD + \bar{A}BCD + \bar{A}BC\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$= m_{15} + m_7 + m_6 + m_9 + m_8 + m_1 + m_0$$

$$Y(A, B, C, D) = \sum m(0, 1, 6, 7, 8, 9, 15)$$

Spiral

In order to work with massive data sets, org., should have the capability to store & process large qty of data.

Algo. can perform cal., data process & reasoning tasks

Q. Express the following func. in std. form

$$F = \bar{y}z + wx\bar{y} + wx\bar{z} + w\bar{x}z + w\bar{x}z$$

	A	B
0	0	0
1	0	0
2	0	0
3	0	0
4	1	1
5	1	1
6	1	1
7	1	1

Product of sum

It is the logical product of 2 or more than 2 sum terms. Basically, it is the AND-ing of OR operated variables.

Ex: $Y = (A+B) \cdot (A+B+C) \cdot (\bar{A} + \bar{B} + \bar{C})$

• POS form is written when func. is 0.

Max term: If a boolean func. contains 'n' variables and the sum term contains all the possible combination of n variables, then the sum term is known as max-term.

$$\text{Nb. of max term} = 2^n \quad (n = \text{no. of variables})$$

Figure 3.1 Data science process model.

in std. $\bar{x}z$

	A	B	C	Max term
0	0	0	0	$(A+B+C)$
1	0	0	1	$A+B+\bar{C}$
2	0	1	0	$A+\bar{B}+C$
3	0	1	1	$A+\bar{B}+\bar{C}$
4	1	0	0	$\bar{A}+B+C$
5	1	0	1	$\bar{A}+B+\bar{C}$
6	1	1	0	$\bar{A}+\bar{B}+C$
7	1	1	1	$\bar{A}+\bar{B}+\bar{C}$

#	A	B	C	y/f
	0	0	0	0
	0	0	1	1
	0	1	0	1
	0	1	1	0
	1	0	0	1
	1	0	1	0
	1	1	0	0
	1	1	1	0

$y = (A+B+C)(A+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C})$
 $(\bar{A}+B+C)(\bar{A}+B+\bar{C})$
 $y(A,B,C) = \prod_m (0, 3, 5, 6, 7)$

Q 2
is the

Q. $F(A,B,C) = (\bar{A}+\bar{B}+\bar{C})(A+C)(\bar{A}+\bar{B})$

all
then

$$\begin{aligned}
 & (\bar{A}+\bar{B}+\bar{C})(A+C) + (B \cdot \bar{B}) \cdot (\bar{A}+\bar{B}) + (C \cdot \bar{C}) \\
 & (\bar{A}+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C}) \\
 & (\bar{A}+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C}) \\
 & \quad m_7 \quad m_0 \quad m_2 \quad m_6 \quad m_7
 \end{aligned}$$

$$y = \prod_m (0, 2, 6, 7)$$

Spiral

work with massive data
org., should have the capability
to store & process large qty
of data.

& reasoning

... of super machine or al

Q. $F(A, B, C, D) = (A + \bar{B} + C)(A + \bar{B})(A + \bar{C} + \bar{D})(\bar{A} + B + C + \bar{D})(B + \bar{C} + \bar{D})$

$= (A + \bar{B} + C)(A + \bar{B})$

$= (\cancel{A + \bar{B} + C + D\bar{D}})(\cancel{A + \bar{B}})$

$= (A + \bar{B} + C) + (D \cdot \bar{D})(A + \bar{B}) + (C \cdot \bar{C})(D \cdot \bar{D})(A + \bar{C} + \bar{D}) + (B \cdot \bar{B})(\bar{A} + B + C + \bar{D})(B + \bar{C} + \bar{D}) \cdot (A \cdot \bar{A})$

$= (\bar{A} + B + C + D)(A + \bar{B} + C + \bar{D})(A + B + \bar{C} + \bar{D})(A + \bar{B} + \bar{C} + \bar{D})$

$(\bar{A} + B + C + \bar{D})(A + B + \bar{C} + \bar{D})(\bar{A} + B + \bar{C} + \bar{D})(A + \bar{B} + C + \bar{D})$

$(A + \bar{B} + \bar{C} + \bar{D})(A + \bar{B} + C + \bar{D})(A + \bar{B} + \bar{C} + C)$

$= \Pi m(3, 4, 5, 6, 7, 9, 11)$

Conversion of SOP to POS :

for 3-Variable

$Y = \sum m(0, 1, 4, 6) \text{ (SOP)}$



$Y = \Pi m(2, 3, 5, 7) \text{ (POS)}$

Q. Con

Q. Convert the following boolean expression

$$Y = A B \bar{C} + \bar{A} B C + A B \bar{C} + \bar{A} \bar{B} \bar{C}$$

$$Y = \sum m(0, 2, 3, 6)$$

$$Y = \prod M(1, 4, 5, 7)$$

Karnaugh map (K-map):

Many times the simplification of boolean func. or expression using boolean laws & theorem becomes complex. So we use K-map. It is the graphical method which is used to simplify a logic eqⁿ or to convert a truth table to its corresponding logic ckt. The no. of cells in K-map = 2^n (n = no. of variables)

Two variable $\Rightarrow$

$$2^2 = 4 \text{ cells}$$

A \ B	0	1
0	m_0	m_1
1	m_2	m_3

Three variable $\Rightarrow$

$$2^3 = 8 \text{ cells}$$

A \ BC	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Spiral

work with massive data sets, orgs, should have the capability to store & process large qty

algo, can cal., data & reason

Four variable $\Rightarrow 2^4 \Rightarrow 16$

AB \ CD	00	01	11	10
00	m_0	m_1	m_2	m_3
01	m_4	m_5	m_6	m_7
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

K-map grouping:

$\uparrow$ 1's
 $\uparrow$ 1's
 $\uparrow$ 1's
 $\uparrow$ 1's
 16

Q. $Y = \sum m(m_0, m_3)$

A \ B	0	1
0	m_0	m_1
1	m_2	m_3

$= \bar{A}B + AB$

A \ B	0	1
0	1	
1	1	

$= B$

A \ B	0	1
0	1	1
1		

$= \bar{A}$

A \ B	0	1
0	1	1
1	0	1

$Y = \bar{A} + B$

$Y = \bar{A} + B$

Q.

A \ BC	00	01	10	11
0	1	0	0	0
1	0	0	0	1

$= \overline{A} \overline{C}$

Q.

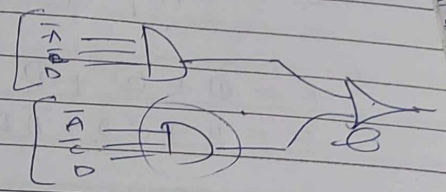
AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

$Y = D$

Q.

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	0	1
10	1	0	0	1

$Y = \overline{D}$



Q. Minimize the boolean expression using K-map.
 $F(A, B, C, D) = \sum m(0, 4, 6, 7, 8, 9, 10, 13)$

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	0	0	1
11	0	1	0	0
10	1	1	0	1

Spiral

In order to work with massive data sets, orgs. should have the capability to store & process large qty of data.

Algo. can be cal., data p. & reasoning

AI is the design of smart machine or al. can perform tasks, which require human

$$Y = \textcircled{1} + \textcircled{2} + \textcircled{3} + \textcircled{4}$$

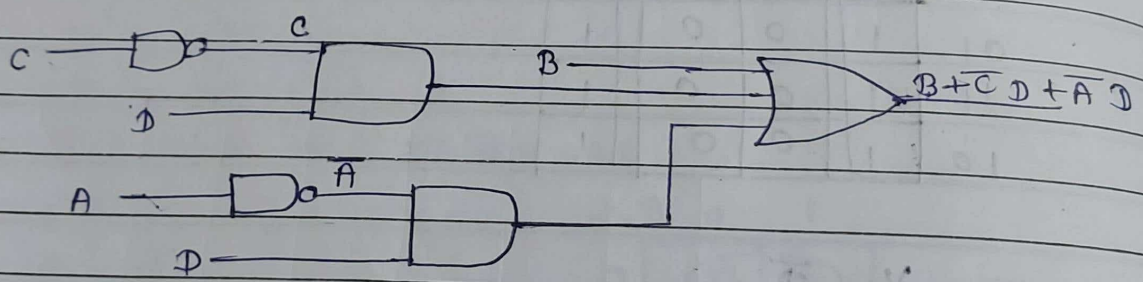
$$Y = \overline{A}\overline{C}D + \overline{A}BC + A\overline{C}D + A\overline{B}D$$

Q. $F(A, B, C, D) = \sum m(1, 3, 4, 5, 6, 7, 9, 12, 13, 14, 15)$ & design with logic gates.

AB \ CD	00	01	11	10
00	0	1	1	3
01	1	4	5	6
11	1	1	1	1
10	8	9	11	10

$$F = \textcircled{1} + \textcircled{2} + \textcircled{3}$$

$$F = B + \overline{C}D + \overline{A}D$$



$$Y = \prod m(0, 2, 8, 10, 11)$$

Q. Design with logic gates

$$Y = \sum (m_0, m_1, m_4, m_5, m_6, m_7, m_8, m_9, m_{10}, m_{11}, m_{12}, m_{13}, m_{14})$$

AB \ CD	00	01	11	10
00	1	1		
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

AB \ CD	00	01	11	10
00	1	1		
01	1	1		
11	1	1		
10	1	1		

$$Y = \bar{C} + \bar{A}B + \bar{A}\bar{B} + A\bar{D}$$

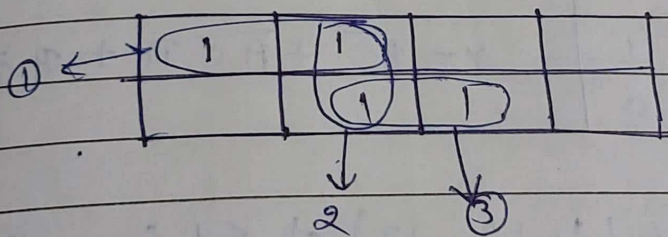
Implicants ÷

It refers to any product / minterm in the SOP or sum/maxterm in POS of boolean funcⁿ.

$$Y = \underbrace{\bar{A}\bar{B}}_{\text{Implicants}} + \underbrace{CD}_{\text{Implicants}} + \underbrace{\bar{A}BC}_{\text{Implicants}}$$

Prime imp ÷

A group of pair or square or rectangles made up of a bunch of adjacent minterm is called prime imp.



$$\text{Prime imp} = 3$$

Essential prime imp ÷

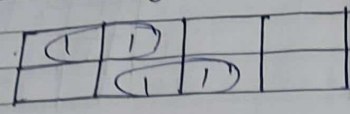
These refer to those groups that cover atleast one of the minterms that

Spiral

In order to work with massive data sets, org^s, should have the capability to store & process large qty of data.

Algo. cal., data & rea

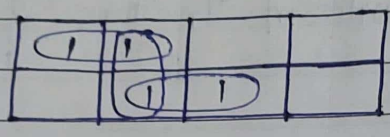
can't ^{get} covered by another prime imp.



= 2

Redundant prime imp.

The RPI refer to the prime implicants for which every one of its minterm gets covered by some other imp. P.I.



Don't care :-

$$f(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$$

AB \ CD	00	01	11	10
00	0	1	0	X
01	X	1	0	1
11	1	1	0	1
10	0	0	0	0

$$Y = B\bar{C} + \bar{A}\bar{C}D + B\bar{D}$$

$$F(A, B, C, D) = \sum m(0, 2, 6, 10, 11, 12, 13) + \sum d(3, 4, 5, 14, 15)$$

AB \ CD	00	01	11	10
00	1		X	1
01	X	X		1
11	1	1	X	X
10			1	1

$$Y = B\bar{C} + AC + \bar{A}\bar{D}$$

Spiral

5 variable ÷

$$Q.f(A, B, C, D, E) = \sum m(0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 13, 15, 16, 18, 20, 21, 22, 23, 24, 25, 26, 27, 29)$$

$$Y = \overline{B} \overline{E} + CE + \overline{B}C$$

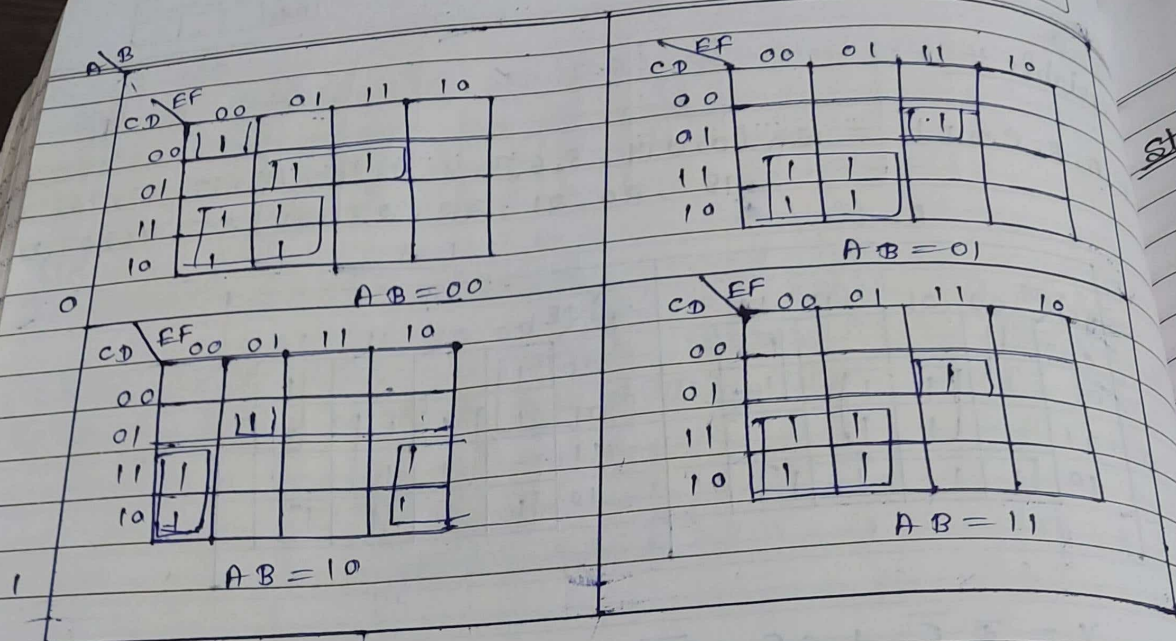
6-variable K-Map:

Q. $F(A, B, C, D, E, F) = \sum_m (0, 5, 7, 8, 9, 12, 13, 23, 24, 25, 28, 29, 37, 40, 42, 44, 46, 0, 55, 56, 57, 60, 61)$

Spira

✓ Algebra

In order to K with massive data sets, K stability



Step-②

$$Y = \overline{A} \overline{C} \overline{E} + B \overline{C} \overline{E} + A \overline{B} \overline{C} \overline{F} + B \overline{C} \overline{D} \overline{E} \overline{F} + \overline{A} \overline{B} \overline{C} \overline{D} \overline{F} + \overline{A} \overline{B} \overline{D} \overline{E} \overline{F} + \overline{B} \overline{C} \overline{D} \overline{E} \overline{F}$$

Quine-mccluskey technique (Tabular method) :-

$$Y(A, B, C, D) = \sum m(0, 1, 3, 7, 8, 9, 11, 15)$$

Step-①	Group	Minterm	Binary Red			
			A	B	C	D
	0	0(m ₀)	0	0	0	0
	1	1	0	0	0	1
		8	1	0	0	0
	2	3	0	0	1	1
		9	1	0	0	1
	3	7	0	1	1	1
		11	1	0	1	1
	4	15	1	1	1	1

Spiral

Step - ②

Group	Matched pair	A	B	C	D
0	0 (no - 1)				
1	0 - 8	0	0	0	
	1 - 3		0	0	0
	1 - 9	0	0		1
	8 - 3		0	0	0
	8 - 9		X		
2	3 - 7	1	0	0	
	3 - 11	0		1	1
	9 - 7		0	1	1
	9 - 11		X		
3	7 - 15		1	0	1
	11 - 15			1	1

accuracy

data
table
ness

Step - ③

Group	Matched pair	A	B	C	D
0	0 - 1 - 8 - 9		0	0	
	0 - 8 - 1 - 9		0	0	
1	1 - 3 - 9 - 11		0	1	
	1 - 9 - 3 - 11		0	1	
2	3 - 7 - 11 - 15			1	1
	3 - 11 - 7 - 15			1	1

sation

dge

Spiral

In order to work with massive data sets, org., should have the capability to store & process large qty of data.

Algo. can cal., data & reason

P, I.	(involved) Minterm	0	1	3	7	8	9	11	15
$\overline{B}\overline{C}$	0, 1, 8, 9	⊗	X			⊗	X		
$\overline{B}D$	1, 3, 9, 11		X	X			X	X	
CD	3, 7, 11, 15			X	⊗			X	⊗

$$E.P.I. = \overline{B}\overline{C} + CD$$